

# Geography Ch-1 — The Earth and the Universe

Physical Geography | Earth in the Solar System | GS-I

FOUNDATION: G.C. Leong · STANDARD: Majid Husain (Indian & World Geography)

Register-style revision notes · grounded in source books

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Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

## FOUNDATION — The Earth, its Shape, Motions & Location (G.C. Leong)

GS-I • Physical  
Geography →  
Earth-Sun  
System (Source:  
G.C. Leong,  
Certificate  
Physical &  
Human  
Geography)

### ■ NEWS TRIGGER

Time-zone and date-line concepts recur in prelims map/logic items; India's single IST (UTC+5:30) vs demands for a separate North-East time zone keep 'longitude & time' exam-relevant.

### ■ INTRO & ORIGIN

The Earth is a near-sphere (an **oblate spheroid / geoid**) — flattened at the poles and bulging at the equator. Leong opens with the proofs of its sphericity and its two basic motions.

*Two motions drive time and seasons: **rotation** (spin on axis → day & night) and **revolution** (orbit round the Sun → the year and seasons).*

### ■ KEY DATA

| Concept    | Key value / rule                        | Consequence                            |
|------------|-----------------------------------------|----------------------------------------|
| Rotation   | ~24 hours; ~1,675 kmph at equator       | Day & night; apparent Sun movement E→W |
| Revolution | 365.25 days around the Sun              | The year; leap-year adjustment         |
| Axial tilt | 23.5° from vertical to orbital plane    | Seasons; changing day length           |
| Latitude   | Angular distance N/S of Equator (0–90°) | Locates place; climate belts           |
| Longitude  | Angular distance E/W of Prime Meridian  | Fixes time; 15° = 1 hour               |
| Date Line  | ≈180° meridian (IDL)                    | Calendar day changes on crossing       |

### ■ STATIC THEORY

- **Evidence of sphericity:** circular horizon, ships' hulls vanishing first, curved shadow in lunar eclipses, and photos from space.
- **Day & night** result from rotation — one half of the globe faces the Sun while the other is dark.
- **Latitude** lines (parallels) run E–W and measure distance from the Equator; key parallels = Tropics (23.5°) and Arctic/Antarctic Circles (66.5°).
- **Longitude** lines (meridians) run N–S from pole to pole; the Prime Meridian (0°) passes through Greenwich.
- **Longitude & time:** Earth turns 360° in 24 h, so 15° of longitude = 1 hour; places east are ahead in time.
- **Standard Time / time zones** use a central meridian for a country (India = 82.5°E → IST = UTC+5:30); the **International Date Line** near 180° separates calendar days.

### ■ MUST-KNOW FACTS

1. Earth's true shape = oblate spheroid / geoid (polar flattening).
2. Rotation ≈ 24 h (day-night); Revolution = 365.25 days (year).

### ■ UPSC TRAPS

- ✗ Rotation causes seasons.
- ✓ Rotation causes day & night; the 23.5° axial TILT + revolution cause seasons.

3. Axial tilt =  $23.5^\circ$  → cause of seasons.

4.  $15^\circ$  longitude = 1 hour of time.

5. Indian Standard Time meridian =  $82.5^\circ\text{E}$  (UTC+5:30).

6. International Date Line  $\approx 180^\circ$  meridian.

✗ Latitude fixes time.

✓ LONGITUDE fixes time ( $15^\circ=1\text{ h}$ ); latitude fixes climate/position N-S.

✗ The Earth is a perfect sphere.

✓ It is an oblate spheroid/geoid — bulged at equator, flattened at poles.

#### ■ MAINS ANGLE

*The longitude-time relationship underpins debates such as a separate North-East India time zone — a crisp application of Earth-Sun geometry to governance.*

#### ■ STUDY LINK

Geography → Physical → Earth as a planet (Leong Ch-1)

HIGH

## STANDARD — Origin of the Universe & Earth's Interior (Majid Husain)

GS-I ·  
Geomorphology  
→ Universe &  
Earth's Interior  
(Source: Majid  
Husain, Indian &  
World  
Geography)

#### ■ NEWS TRIGGER

Seismic-wave study of the Earth's interior (P/S wave behaviour) is the same physics used to locate earthquake foci — directly linked to India's high-seismicity Himalayan zones.

#### ■ INTRO & ORIGIN

The **Big Bang** (~15 billion years ago) caused matter to expand into billions of galaxies, stars and planets. Our **solar system** formed by the gravitational collapse of a cold cloud of gas and dust (mainly hydrogen & helium) in a spiral arm of the Milky Way.

*Planetary composition varies with distance from the Sun: inner **terrestrial planets** formed from rock/metal that crystallised at high temperature; outer planets from materials condensing at low temperature. A **galaxy** = huge mass of stars + nebulae; a **nebula** = cloud of dust & gas within it.*

#### ■ KEY DATA

| Layer      | Extent / depth               | Composition & state                 |
|------------|------------------------------|-------------------------------------|
| Crust      | Surface to Moho (~35 km avg) | Rigid; <1% of Earth's volume        |
| Mantle     | Moho (~35 km) → 2900 km      | Solid rock + magma; density 3.3–5.7 |
| Outer core | 2900 km → ~5150 km           | Liquid nickel-iron (NiFe)           |
| Inner core | ~5150 km → centre            | Solid NiFe; ~2700°C+                |

#### ■ STATIC THEORY

- **Mohorovicic Discontinuity (Moho)** separates crust from mantle (~35 km average depth).
- **Gutenberg Discontinuity** (~2900 km) separates the mantle from the outer core.
- **Seismic evidence:** S-waves (shear) cannot pass liquids and vanish at 2900 km → the outer core is LIQUID; P-waves pass through all layers.
- P/S wave velocities reveal: solid inner core, liquid outer core, thick mantle, soft asthenosphere, rigid lithosphere.
- The **core** is a dense nickel-iron ('nife') alloy; density and temperature rise sharply with depth.

- **Axial parallelism:** Earth's axis stays fixed (pointing at Polaris) through its orbit — combined with the 23.5° tilt this creates the seasonal march of the Sun.

#### ■ MUST-KNOW FACTS

1. Big Bang ≈ 15 billion years ago; solar system by gravitational collapse of gas-dust cloud.
2. Moho = crust/mantle boundary (~35 km); Gutenberg = mantle/outer-core (~2900 km).
3. S-waves disappear at 2900 km → LIQUID outer core (key seismic proof).
4. Core = nickel-iron (nife); outer liquid, inner solid.
5. Mantle density 3.3–5.7; crust < 1% of Earth's volume.
6. Terrestrial (inner) planets = rock/metal; Jovian (outer) = gas/ice condensates.

#### ■ UPSC TRAPS

- ✗ The whole core is molten liquid.
- ✓ Only the OUTER core is liquid; the inner core is solid (confirmed by seismic-wave behaviour).
- ✗ Moho separates mantle from core.
- ✓ Moho = crust/mantle boundary; the mantle/outer-core boundary is the Gutenberg discontinuity.
- ✗ P-waves cannot travel through liquids.
- ✓ S-waves cannot pass liquids; P-waves travel through solids AND liquids.

#### ■ MAINS ANGLE

*Explain how seismic-wave (P/S) behaviour maps the Earth's layered, differentiated interior — the foundation for understanding plate tectonics, volcanism and India's seismic hazard.*

#### ■ STUDY LINK

Geography → Geomorphology → Universe & Earth's Interior (Majid Husain)

End of Register Notes — UPSC Agent / Copilot CLI